FAYE (FEI) XIA

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CMU Chemical Engineering student and AI startup co-founder experienced in translating user needs into technical requirements and coordinating stakeholders and engineers across time zones. Applies engineering judgment to evaluate AI capabilities and tradeoffs, select suitable approaches, and turn product ideas into practical solutions.

EDUCATION

Carnegie Mellon University Expected May 2028
B.S. Chemical Engineering; Minors in Biomedical Engineering and Machine Learning

EXPERIENCE

Mafold July 2026 - Present
Co-founder | AI Agent Product Development | Website
- Identified a gap in sharing personal-agent work across teams; proposed agent-to-agent collaboration and partnered with a backend engineer to break the concept into technical tasks and privacy requirements.
- Co-defined an access-control approach using owner-managed allowlists and owner-consent escalation for sensitive requests, addressing authorization and information-sharing risks in the proposed workflow.
- Helped establish weekly team syncs and shared issue tracking in group chat across a 12-hour time difference; defined agent-first troubleshooting and backend escalation to address duplicate work and engineer workload.

Procter & Gamble | Oral Care R&D May 2026 - August 2026
Digital Innovation Intern | Portfolio
- Gathered stakeholder needs for an R&D knowledge chatbot, clarified expectations with a manager and coach, and translated user requests into technical questions and requirements for engineering discussions.
- Coordinated meetings with a Europe-based engineer and scheduled user testing to collect feedback before subsequent engineering discussions, connecting stakeholder needs with technical iteration.
- Evaluated an existing RAG pipeline against R&D retrieval needs; identified mismatches in query interpretation and retrieval relevance, informing the choice of Databricks-native retrieval and text-to-SQL tools.
- Built an automated data pipeline across 5,000+ research documents, parsing irregular formats and embedded attachments into Databricks to support chatbot access to R&D information.

Carnegie Mellon University | Kitchin Lab December 2025 - Present
Researcher | Opentrons code | Viscosity research
- Audited a viscosity ML benchmark spanning 39,077 measurements; standardized 1,733 molecular representations into 1,367 unique molecules, uncovered training overlap in 27 of 65 test molecules, and designed 2 evaluation protocols to assess generalization.
- Designed and executed 8 robustness tests for MixEmbed, identifying 7 defects across input validation, missing-data handling, and dependency compatibility; communicated findings to the library maintainer, resulting in fixes for all 7 issues.
- Developed Opentrons Flex protocols for 96 dye mixtures at 4 wavelengths; validated predictive models on 20 held-out wells (R-squared: 0.9912-0.9989).

PRODUCT & PROJECT LEADERSHIP

L'Oreal Brandstorm | ECHOlogne 2026
Fragrance Innovation Competition | Initial-stage participant | Project
- Conducted luxury-fragrance market and competitor research to inform Prada brand selection and product positioning; identified opportunities in personalization and differentiation from lower-cost alternatives.
- Helped define features, pricing, and a phased rollout roadmap for a proposed refillable fragrance system with an NFC-connected app, aligning the product concept with Prada brand identity.

Monument Valley Game Replica | CMU 15-112 May 2025
Group Leader | GitHub | Video
- Led a student team in developing a playable Python game integrating user interaction and game logic; selected for the Top 20 Projects Gallery out of 150 projects.

SKILLS & HONORS

Program & product: Technical requirements, task breakdown, cross-functional coordination, risk assessment, issue prioritization, user testing, product roadmapping.
Technical: RAG, LLM orchestration, SolidWorks, COMSOL Multiphysics, Power BI, KNIME, Python (NumPy, pandas, scikit-learn), SQL, C/C++, MATLAB, Git, Databricks, text-to-SQL, regression, cross-validation, Opentrons Flex.
Honors: Dean's List (Fall 2024, Spring 2026); $4,500 ChESS Research Fellowship (2025); patent co-inventor (2024).